

Link to repository: [Ojas2011/Telco-Customer-Churn-Prediction](https://github.com/Ojas2011/Telco-Customer-Churn-Prediction)

Introduction:

In a company called Telco, there are many customers who have different kinds of feedback to give on different things. And through that feedback we can see what kinds of things the organization is lacking and what kinds of things the organization needs to work on, but what if we could predict based on the feedback whether a customer would return or not given their response to different questions? Therefore, my research question was that: based on customer reviews and their responses to different questions about Telco, what is the likelihood of them returning as customers again? Through this we will answer and predict how customer feedback is connected to customer churn.

Data:

The data used in this project was found from the github repository containing the dataset 'Telco-Customer-Churn'. It has 7043 rows which show the 7043 different customers and their different opinions/satisfaction with the services, hence showing a diverse dataset with customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Churn.

Below is a formal description of the variables(taken from the github repository):

- customerID: ID of specific customers
- gender: Gender of customer
- SeniorCitizen: Whether the customer is a senior citizen or not
- Partner: Whether the customer has a partner or not
- Dependents: Whether the customer is a dependent or not
- tenure: Tenure of a customer
- PhoneService: Which phone service customer uses(if they use it)
- MultipleLines: Whether customer has multiple lines
- InternetService: Which internet service(if used) is used by a customer
- OnlineSecurity: Whether customer has online security
- OnlineBackup: Whether customer has online backup
- DeviceProtection: Whether customer has device protection
- TechSupport: Whether customer has tech support
- StreamingTV: Whether customer has/uses streaming TV
- StreamingMovies: Whether customer has/uses streaming movies
- Contract: Description of customer contract
- PaperlessBilling: Whether customer has paperless billing
- PaymentMethod: Payment method used by the customer
- MonthlyCharges: Monthly charges for a customer
- TotalCharges: Total amount of charges for a customer

- Churn: Whether customers want to return or not

The dependent variable of this dataset is churn, which has two unique variables “No” which is in 5174 rows for no returning of customer and “Yes” for returning of customer which is in 1869 rows.

Preprocessing and Exploratory Data Analysis:

After reviewing the initial dataset and coding it on Visual Studio Code, there were no null values present. Because there is a huge split on customers saying “Yes” and customers saying “No” (People saying yes in 1000s and people saying no in 5000s) there had to be some cleaning done to keep it unbiased.

Initial Exploratory Data analysis showed an imbalance. The dataset contains 5174 customers saying they wouldn't like to continue with Telco and 1869 customers saying they would like to continue. This shows that the statistics may not fully be unbiased, however later on we did get back a decision tree accuracy of 77.11 percent.

There are also other biases with a significant split between other variables such as online security where 3498 have no online security, 2019 do and 1526 have no internet in the first place which would cause bias since online security is a variable to be taken into account, but if someone does not have internet in the first place what answer can they give, there are many such biases in this dataset.

Due to such biases, it is fair to say that using additional things such as the statistical descriptions for our dataset, the correlation heatmap which shows performance based on the relations between different variables.

Modeling:

To ensure we minimize the bias and maximize the accuracy, we cleaned the variables/responses in places where they were not necessary such as people having no internet services were out of the dataset for things such online security and tech support variables since those not having internet in the first place can't really help with questions about tech support. We also had to remove the customer ID everywhere since it had nothing to do with customer churn or any other variable. While tenure was a minor variable and therefore the data for it was not cleaned it is still an important factor since the amount of years a specific person was with a company impacted their trust in it and their willingness to stay with it regardless of latest updates.

Out of the many models used, some important ones were RandomForestClassifier, DecisionTreeClassifier, LabelEncoder, classification_report. Performance was evaluated using the feature importances used in Random Forest, the accuracy data contained in the Logistic Regression and the precision, recall, f1-score, support, churn/no churn, accuracy, macro average, weighted average in the decision tree accuracy.

Logistic Regression was used for helping categorize the different variables where they belonged based on how it was supposed to be managed statistically. Random Forest was there to make predictions for our dataset and show that monthly charges, yearly charges and tenure were relatively more significant factors used to determine/predict which customers are likely to return. Decision Trees were there to help organize all the possibilities and give their statistical possibility based on many combinations of different possible outcomes.

Results:

Of the models mentioned, Random Forest had an accuracy of 0.7861, Decision Tree had an accuracy of 0.7164, Logistic Regression had an accuracy of 0.7939 which means that Logistic Regression performed better than the other two. The reason logistic regression did better most likely has to do with fact that type of learning it used with the dataset. This shows that the model did a relatively decent job of seeing how different variables impacted customer churn for Telco.

When seeing what impacts customer churn, monthlyCharges and totalCharges play a good role, including but not limited to the contract and tenure in which higher charges will impact whether the customer wants to return or not as well as how long their tenure and contract is.

The feature importance chart that uses random forest shows how charges and tenure are having an impact more significantly than something like age, marital status and gender and some variables which are not even considered over here due to them not having any real meaning in this dataset.

Conclusion:

Limitations and future work: In the real world, such a dataset would not exactly be considered perfect(not that it ever is most of the time) or good enough but comparatively it would be considered decent since all the accuracy percentages are in the 70s. However because more people said no and all had different answers we can't assume that all people who have been associated as customers with Telco want a higher monthly price but lower total price or a higher tenure will lead them to want to stay, it is important to take into account that services such as tech support and internet being good or bad cannot be counted for everyone since some people don't have internet or the necessary technology with them in the first place, something that Telco has nothing to do with, and it's variables like these that should've been taken into consideration before making any decision or showing accuracy or any statistical values.

Overall the models and the decision tree were more or less the same when it came to performance, with the decision tree performing slightly lower. A limitation could be the certain variables and their impact on the values of the models for both the decision tree and the other models. For future reference it will be good to do a lot more data cleaning to have better

accuracy, there could be different models than the one used here to see possible improvements or things missed out here.

```
Logistic Regression: Accuracy = 0.7939
Decision Tree: Accuracy = 0.7164
Random Forest: Accuracy = 0.7861
Gradient Boosting: Accuracy = 0.7967
AdaBoost: Accuracy = 0.7868
Extra Trees: Accuracy = 0.7704
SVC: Accuracy = 0.7854
KNN: Accuracy = 0.7484
Naive Bayes: Accuracy = 0.7328
LDA: Accuracy = 0.7960
MLP: Accuracy = 0.7754
```

Best model: Gradient Boosting with accuracy 0.7967

```
Decision Tree Accuracy: 0.7711
precision    recall  f1-score   support

   No Churn    0.81    0.90    0.85    1033
   Churn       0.60    0.41    0.49     374

 accuracy      0.77    1407
 macro avg     0.71    0.66    0.67    1407
weighted avg     0.75    0.77    0.76    1407
```

Resources:

Nas-virat. (n.d.). *Telco-customer-churn* [Dataset]. GitHub.

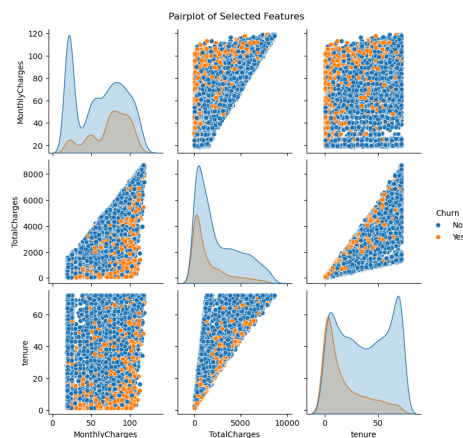
<https://github.com/Nas-virat/Telco-Customer-Churn>

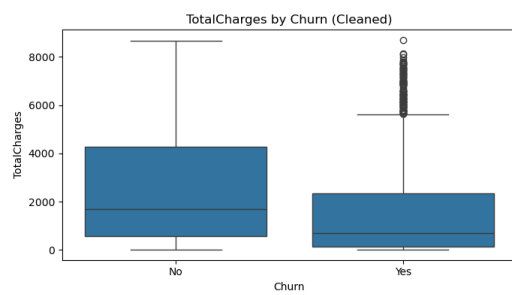
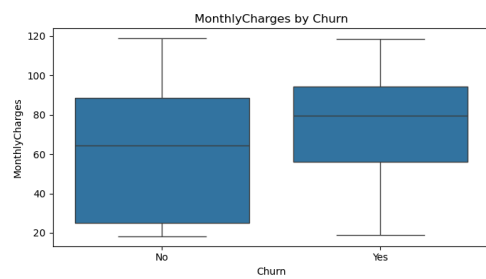
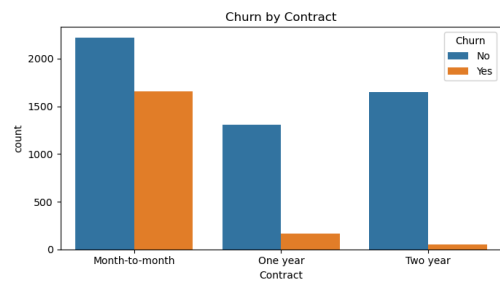
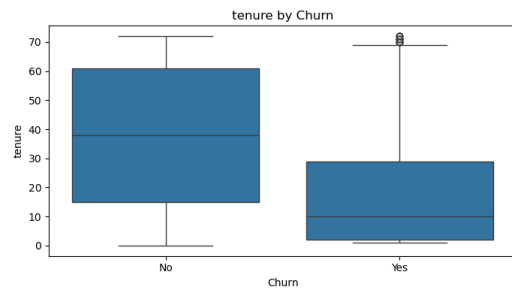
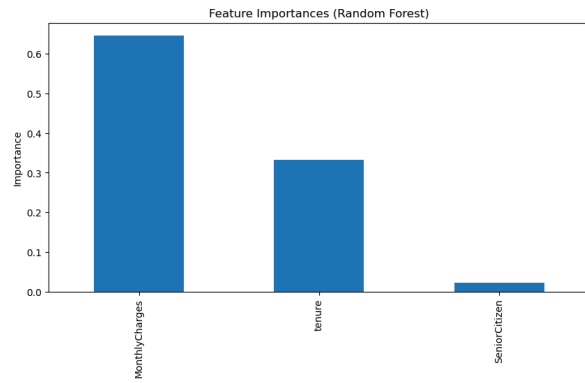
- I also used the scikit learn part 2 Jupyter Notebook code for reference on some parts of the code.

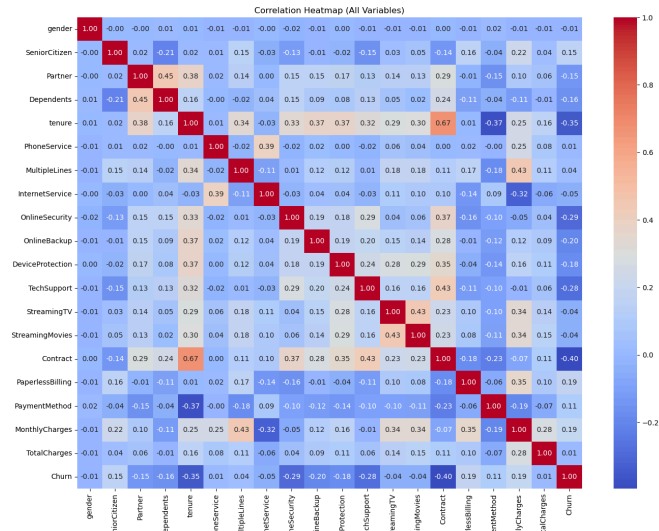
AI Acknowledgement:

- Github Copilot was used to bring all my ideas for visualizations and statistics to write the code for both.

Appendix:







	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Decision Tree Accuracy: 0.7711				
	precision	recall	f1-score	support
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accuracy			0.77	1407
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Decision Tree for Churn Prediction

